

Course Policy of
Chemical Reaction Engineering II (CL 24303)
Session: MO2026 Section: A

Instructor:

Sangram Roy (sroy@bitmesra.ac.in/4366)

Textbooks:

OL- Chemical Reaction Engineering, 3rd Edition, By Octave Levenspiel

HSF – Elements of Chemical Reaction Engineering, 5th Edition, By H. Scott Fogler

Reference:

JJC- Chemical and Catalytic Reaction Engineering, By James J. Carberry

JMS – Chemical Kinetics, 2nd Edition, By J.M. Smith

Evaluation:

Assessment	Marks	Module
Quiz 1	10	1
Mid Sem	25	1, 2
Viva	10	All
End Sem	50	All
Tutorials	5	All
Total	100	

Tutorials:

- The tutorials are designed for problem-solving and explanation of key concepts pertaining to the course
- Although discussions among peers are strongly encouraged, they are to be worked out independently
- It is mandatory that completed tutorials are submitted within the assigned deadline
- The aggregate marks of tutorials will be scaled to 5 under assessment

Viva:

- In lieu of Quiz 2, a viva will be conducted on the tutorials and design problem at the end of the course

Class Timings:

Monday (15:30-16:20, Room G1)

Tuesday (11:00-11:50, Room 217)

Wednesday (16:30-17:20, Room G1)

Friday (15:30-16:20, Room - FIII)

Extra classes, if required

Availability after class: Thursday afternoons or through e-mail

Attendance Policy:

It is expected that everyone will have 100% attendance. Any consideration due to a shortfall is subject to the institute's rules. Medical exigencies will be accepted only if a valid certificate is provided for three consecutive absences (even if there is a weekend or a holiday in between).

Lecture Plan for Chemical Reaction Engineering II (CL 24303) in MO2026

L No.	Date	Time	Location	Module	Topics	Reference	Remarks
1	27-07-2026	15:30-16:20	G1	-	Introduction, Course Policy, Logistics, Syllabus		
2	28-07-2026	11:00-11:50	217	0	Revision of CRE- I Rate equation, Types of reactors	JJC-1-1, 2-1,2-2, 2-3 OL – 2.1, 2.2, 3,4,5	
3	29-07-2026	16:30-17:20	G1	0	Revision of CRE- I Rate equation, Types of reactors	JJC-1-1, 2-1,2-2, 2-3 OL – 2.1, 2.2, 3,4,5	
4	31-07-2026	15:30-16:20	FIII	0	Tutorial - 0		
5	03-08-2026	15:30-16:20	G1	1	Introduction/Classification/ Heterogenous Reaction	JMS – Ch7 JJC – 5-1,5-2,5-3	
6	04-08-2026	11:00-11:50	217	1	Concept of Intrinsic and global rate of reaction – Mass transfer effects	JJC-1-1, 2-1,2 HSF – Ch14	
7	05-08-2026	16:30-17:20	G1	1	Catalyst Characterization	JMS – 8-4,8-5,8-6, 8-7	
8	07-08-2026	15:30-16:20	FIII	1	Catalyst Characterization	JMS – 8-9, 8-10	
9	10-08-2026	15:30-16:20	G1	1	Catalyst Characterization	JMS – 8-4,8-5,8-6, 8-7	
10	11-08-2026	11:00-11:50	217	1	Tutorial - 1		
11	12-08-2026	16:30-17:20	G1	2	Steps in heterogeneous reaction, Adsorption	HSF – 10.1, 10.2	
12	14-08-2026	15:30-16:20	FIII	2	Adsorption Isotherms	JMS – 8-4	
13	17-08-2026	15:30-16:20	G1	2	Adsorption Isotherms, Surface reaction/ Desorption	JJC – 8-9, 8-10, 8-11	
14	18-08-2026	11:00-11:50	217	2	Adsorption Isotherms, Surface reaction/ Desorption	JJC – 8-9, 8-10, 8-11	
15	19-08-2026	16:30-17:20	G1	2	Tutorial -2		

16	21-08-2026	15:30-16:20	FIII	2	Deduction of rate law, rate-limiting step	HSF - 10.3 JMS – 9-3, 9-4, 9-5	
17	24-08-2026	15:30-16:20	G1	2	Deduction of rate law, rate-limiting step	HSF - 10.3 JMS – 9-3, 9-4, 9-5	Quiz 1
18	25-08-2026	11:00-11:50	217	2	Deduction of rate law, rate-limiting step	HSF - 10.3 JMS – 9-3, 9-4, 9-5	
19	31-08-2026	15:30-16:20	G1	2	Tutorial - 3		
20	01-09-2026	11:00-11:50	217	3	Kinetics of solid-catalysed gas-phase reaction	OL- Ch18 HSF – Ch 15	
21	02-09-2026	16:30-17:20	G1	3	Isothermal inter- and intraphase effectiveness factors	OL- Ch18 HSF – Ch 15	
22	07-09-2026	15:30-16:20	G1	3	Isothermal inter- and intraphase effectiveness factors	OL- Ch18 HSF – Ch 15	
23	08-09-2026	11:00-11:50	217	3	Tutorial - 4		
24	09-09-2026	16:30-17:20	G1	3	Isothermal inter- and intraphase effectiveness factors	OL- 19,20, 22	
25	11-09-2026	15:30-16:20	FIII	3	Isothermal inter- and intraphase effectiveness factors	OL- 19,20, 22	Mid-Sem
26	25-09-2026	15:30-16:20	FIII	4	Catalyst Deactivation	HSF – Ch 10	
27	28-09-2026	15:30-16:20	G1	4	Tutorial - 5		
28	29-09-2026	11:00-11:50	217	4	Non-catalytic gas-solid Reaction - examples	OL – Ch 23,24	
29	30-09-2026	16:30-17:20	G1	4	Models of non-catalytic gas-solid reaction – Shrinking Core Model (SCM)	OL – Ch 25	
30	05-10-2026	15:30-16:20	G1	4	Models of non-catalytic gas-solid reaction – Shrinking Core Model (SCM)	OL – Ch 25	
31	06-10-2026	11:00-11:50	217	4	Models of non-catalytic gas-solid reaction – Progressive Conversion Model (PCM)	OL – Ch 25	
32	07-10-2026	16:30-17:20	G1	4	Tutorial - 6		
33	09-10-2026	15:30-16:20	FIII	5	Design of catalytic gas-solid reactors	OL- Ch 26	

34	12-10-2026	15:30-16:20	G1	5	Design of gas-solid reactors	OL- Ch 26	
35	13-10-2026	11:00-11:50	217	5	Design of gas-solid reactors	OL- Ch 26	
36	14-10-2026	16:30-17:20	G1	5	Tutorial - 7		
37	16-10-2026	15:30-16:20	FIII	5	Gas-liquid reactions	OL – Ch23,24	
38	23-10-2026	15:30-16:20	FIII	5	Gas-liquid reactions	OL – Ch23,24	
39	26-10-2026	15:30-16:20	G1	5	Enhancement factor in gas-liquid reactions	OL – Ch24	
40	27-10-2026	11:00-11:50	217	5	Enhancement factor in gas-liquid reactions	OL – Ch24	
41	28-10-2026	16:30-17:20	G1		Tutorial - 8		
42	30-10-2026	15:30-16:20	FIII		Gas-liquid reactors	OL – Ch24	
43	02-11-2026	15:30-16:20	G1		Tutorial - 9		
44	03-11-2026	11:00-11:50	217		Buffer		
45	04-11-2026	16:30-17:20	G1		VIVA		
46	06-11-2026	15:30-16:20	FIII		VIVA		
47	10-11-2026	11:00-11:50	217		VIVA		Last Class